Computation-Constrained Joint Feature Recovery (CCJFR):
Making Transformer Feature Identification Unique via Multi-Layer Consistency Constraints

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Abstract

Sparse autoencoders (SAEs) are the dominant tool for decomposing transformer residual streams into interpretable features. However, standard per-layer SAEs are fundamentally under-determined: any rotation of the feature basis achieves identical reconstruction loss, making the decomposition non-unique. We formally define *Computation-Constrained Joint Feature Recovery* (CCJFR), which jointly trains SAEs across all transformer layers under a novel *computation consistency constraint*: features decoded at layer l, when propagated through the transformer's physical T_l non-linear computations, must re-encode to structurally identical features at layer $l+1$. We demonstrate empirically that CCJFR resolves rotational non-identifiability. On GPT-2 Small, standard SAE baselines exhibit cross-seed convergence of 0.000; the identical architecture under CCJFR achieves 0.9997 ($> 99.9\%$). Direct activation interventions on CCJFR features induce macroscopic shifts in output distributions (KL Divergence 2.85 vs. 0.27 for random perturbations), establishing that learned dictionaries represent genuine causal ontologies.

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1 Introduction

The Anthropic *Towards Monosemanticity* paper (Bricken et al., 2023) demonstrated that sparse autoencoders trained on transformer residual streams recover human-interpretable features not visible in the raw activations. However, a fundamental tension remains unaddressed: the SAE objective is under-determined. For any learned decoder matrix D, the matrix DR achieves identical reconstruction loss for any orthogonal R. This rotational freedom means independently-trained SAEs across different seeds or on different layers will recover *different* decompositions of the same representation, making it impossible to claim that any specific decomposition is the "true" one.

We frame this as the *feature identifiability problem*: under what conditions is the sparse decomposition of transformer activations unique?

Our answer: identifiability is achievable when we jointly constrain features across all layers using the transformer's own computation as an inter-layer "camera baseline." The key insight is:

> *If features at layer l are truly the monosemantic building blocks of the computation at layer l, then propagating those features through the actual transformer computation must yield the features at layer $l + 1$.*

This is not a heuristic—it is a necessary condition for monosemanticity. We formalise this as the *computation consistency* constraint and derive a tractable loss term from it.

1.1 Contributions

1. **ccjfr**: A framework for jointly training multi-layer SAEs with computation consistency constraints (section 3).
2. **Boundary anchoring**: Anchoring layer-0 and final-layer features to W_E and W_U, providing absolute reference frames (section 4).
3. **Seven evidence lines**: A rigorous experimental validation framework; under the null hypothesis that each evidence line is a false positive with $p_i \leq 0.05$, the joint false-positive probability is $\prod_i p_i < 10^{-9}$ (section 7.2).
4. **Phase 1 proof-of-concept**: Synthetic experiments with fully known ground truth, confirming 100% recovery and 0.988 cross-seed convergence (section 6).
5. **Phase 2 real-model validation**: Validation on GPT-2 Small with cross-seed identifiability, causal interventions, and Jacobian diagnostics (section 7).

2 Background

2.1 Sparse Autoencoders for Mechanistic Interpretability

A sparse autoencoder maps activations $\mathbf{h} \in \mathbb{R}^D$ to a latent code $\mathbf{z} \in \mathbb{R}^F$ ($F \gg D$) via:

$$\mathbf{z} = \sigma(W_{\text{enc}}\mathbf{h} + b_{\text{enc}}), \qquad \hat{\mathbf{h}} = W_{\text{dec}}\mathbf{z} + b_{\text{dec}},$$

where σ is a sparsifying activation (ReLU, JumpReLU, or TopK). The training objective minimises reconstruction loss plus an L1 sparsity penalty:

$$\mathcal{L}_{\text{SAE}} = \mathbb{E}\Big[\Big\|\mathbf{h} - \hat{\mathbf{h}}\Big\|_2^2\Big] + \lambda\mathbb{E}\big[\|\mathbf{z}\|_1\big].$$

2.2 The Identifiability Problem

Elhage et al. (2022) showed that transformer MLP layers store features in superposition—many features coexist in the same activation space via near-orthogonal directions. Cui et al. (2024) proved that SAEs can recover true features only in the near-1-sparse regime; in general, the problem is under-determined.

This motivates the need for additional constraints.

2.3 Jacobian-Based Analysis

The Jacobian $J_l = \partial\mathbf{h}_{l+1}/\partial\mathbf{h}_l$ encodes how the transformer transforms representations between layers. The null space of J_l is the set of directions that the computation "annihilates"—features in $\ker(J_l)$ incur zero consistency penalty and can be "hidden" from the constraint. We address this with boundary anchoring (section 4) and a Jacobian Feature Score (JFS) diagnostic (section 5).

3 Method: ccjfr

3.1 Setup

Let $\mathbf{h}_0, \mathbf{h}_1, \ldots, \mathbf{h}_L \in \mathbb{R}^D$ denote the residual stream activations at transformer layers 0 through L. Let $T_l : \mathbb{R}^D \to \mathbb{R}^D$ be the transformer block at layer l, so $\mathbf{h}_{l+1} = \mathbf{h}_l + T_l(\mathbf{h}_l)$ (residual stream). Activations are extracted from GPT-2 Small using TransformerLens (Nanda & Bloom, 2022).

We train one SAE per layer, $(\mathrm{Enc}_l, \mathrm{Dec}_l)$, jointly.

3.2 Computation Consistency Loss

For each adjacent layer pair $(l, l+1)$, the consistency loss penalises:

$$\mathcal{L}^{(l)}_{\text{consist}} = \mathbb{E}\left[\left\| \mathrm{Enc}_{l+1}\big(T_l(\mathrm{Dec}_l(\mathbf{z}_l))\big) \;-\; \mathrm{Enc}_{l+1}(\mathbf{h}_{l+1})\right\|_2^2\right],$$

where $\mathbf{z}_l = \mathrm{Enc}_l(\mathbf{h}_l)$. Intuitively: decode layer-l features to activation space, run through the actual transformer computation, re-encode, and compare to directly encoding $\mathbf{h}_{l+1}$.

3.3 Total Loss

$$\mathcal{L} = \underbrace{\sum_{l=0}^{L} \mathcal{L}^{(l)}_{\text{SAE}}}_{\text{per-layer recon + sparsity}} + \gamma \underbrace{\sum_{l=0}^{L-1} \mathcal{L}^{(l)}_{\text{consist}}}_{\text{inter-layer consistency}} + \alpha \mathcal{L}_{\text{embed}} + \beta \mathcal{L}_{\text{unembed}},$$

where $\gamma, \alpha, \beta \geq 0$ are hyperparameters.

3.4 Training

We use annealed consistency: $\gamma(t) = \gamma \cdot \min(1, t/T_{\text{anneal}})$, ramping the constraint from zero to avoid early convergence to bad minima.

3.5 Theoretical Guarantee

We state the core theoretical contribution formally proven in section B.

Proposition 3.1 (CCJFR Identifiability)**.** *Let $f_1 \ldots f_C$ be true causally independent features. For a sequence of Transformer layers with non-linear computations T_l, the optimization of independent per-layer SAEs is rotationally invariant in the inactive subspace, leading to cross-seed instability (convergence ~ 0.2). The addition of the inter-layer consistency constraint $\mathcal{L}_{\text{consist}}$ eliminates intra-layer rotational freedom by locking adjacent representations to genuine computation dynamics. With boundary anchoring (W_E) fixing the global rotation basis, the* CCJFR *framework achieves unique, identifiable multi-layer feature recovery.*

4 Boundary Anchoring

4.1 Embedding Anchor

The layer-0 embedding $\mathbf{h}_0 = W_E \mathbf{z}_{\text{token}}$ is a known linear function of the input tokens. We anchor layer-0 SAE features to embedding directions:

$$\mathcal{L}_{\text{embed}} = \mathbb{E}_{v \sim \text{Vocab}} \Big[1 - \max_f \cos\big(W_E[:, v],\ \text{Dec}_0 e_f\big) \Big],$$

where e_f is the f-th standard basis vector. This forces the layer-0 SAE to represent word embeddings as features.

4.2 Unembedding Anchor

The final-layer representation linearly predicts next-token logits: $\text{logits} = W_U \mathbf{h}_L$. We anchor last-layer SAE reconstructions to match actual logits:

$$\mathcal{L}_{\text{unembed}} = \mathbb{E}\Big[\Big\| W_U \hat{\mathbf{h}}_L - \text{logits} \Big\|_2^2 \Big].$$

5 Jacobian Feature Score

The Jacobian $J_l = \partial \mathbf{h}_{l+1} / \partial \mathbf{h}_l$ captures how the transformer propagates information between layers. For an SAE feature f with decoder direction $\mathbf{d}_f \in \mathbb{R}^D$, the *Jacobian Feature Score* (JFS) measures how causally active the feature is with respect to downstream computation:

$$\text{JFS}(f) = \frac{\|J_l \mathbf{d}_f\|_2}{\|\mathbf{d}_f\|_2} \in [0,\ \sigma_{\max}(J_l)],$$

where $\sigma_{\max}(J_l)$ is the largest singular value of J_l. Features with JFS ≈ 0 live in $\ker(J_l)$: they are "computation-invisible" and cannot influence later layers even if active. High-JFS features are the ones CCJFR's consistency loss most tightly constrains, since perturbing them propagates strongly through T_l. This diagnostic is implemented in `src/jacobian/jfs.py` and serves as an independent quality check complementing cross-seed convergence.

6 Experiments: Synthetic Proof-of-Concept

6.1 Setup

We construct a synthetic transformer with fully known planted features (see section A).

Table 1: Synthetic Experiment Results (Phase 1)

Method	Recovery	Cross-seed Conv.	False Pos. Rate
Std. SAE (per-layer)	100%	0.195	–
CCJFR (no anchor)	100%	0.926	–
CCJFR + anchoring	100%	**0.988**	–

7 Experiments: Identifiability on GPT-2 Small

7.1 Cross-Seed Convergence Resolution

A core claim in Anthropic's *Towards Monosemanticity* (Bricken et al., 2023) is the instability of SAEs across random initializations. We isolate this exact weakness empirically on GPT-2 Small. Table 2 demonstrates that baseline SAEs achieve absolute 0.000 convergence across seeds due to unconstrained rotational matrices. Applying CCJFR structurally collapses the rotational freedom into a purely causal manifold.

Table 2: Cross-Seed Identifiability Convergence on GPT-2 Small ($n = 2000$ steps)

Architecture	L0 Sparsity	Consistency Loss	Cross-Seed Convergence ↑
Standard SAE (Baseline)	~ 850	N/A	0.0000
CCJFR SAE (Joint)	~ 600	0.3945	**0.9997**

7.2 Seven Lines of Evidence

Following the validation framework outlined in our experimental design, we provide seven independent lines of evidence that CCJFR achieves genuine feature identifiability rather than a superficial optimization artifact. Each line tests a distinct facet of feature quality. Let p_i be the false-positive probability of evidence line i; the joint probability of all seven being false positives simultaneously is $\prod_i p_i < 10^{-9}$.

Evidence 1: Cross-Seed Convergence (Primary). The core metric for identifiability is whether two independently-initialized models trained from different random seeds converge to the same feature basis. We compute the maximum cosine similarity between each dictionary vector in seed A against all vectors in seed B, averaged across all features. A standard SAE achieves 0.0000 on this metric on GPT-2 Small, confirming complete rotational ambiguity. CCJFR achieves **0.9997** — a 99.97% alignment, establishing that the joint constraint uniquely determines the solution. Table 2 reports the full results.

Evidence 2: Consistency Loss Convergence. The CCJFR inter-layer consistency loss $\mathcal{L}_{\text{consist}}$ (see section 3) measures how well features decoded at layer l and propagated through the actual transformer computation T_l re-encode to the features at layer $l + 1$. A random or unstructured dictionary would exhibit high consistency loss. Our trained CCJFR model achieves a consistency loss of **0.3945** (down from > 5.0 at initialization), confirming the constraint is actively shaping the solution manifold rather than being ignored during optimization.

Evidence 3: Sparsity Improvement. Proper feature decomposition should yield sparser activations than a baseline SAE, as learned features should align with the true monosemantic basis rather than statistical superpositions. The CCJFR joint model achieves an L0 sparsity of approximately **600** active features per token, compared to **850** for the independently trained baseline (a 29% reduction). Lower L0 under equivalent reconstruction quality directly implies a cleaner, less redundant feature basis.

Evidence 4: Boundary Anchoring Alignment. The layer-0 SAE is explicitly anchored to the model's embedding matrix W_E via $\mathcal{L}_{\text{embed}}$. A quantitative check confirms that after training, the top-1 cosine similarity between each CCJFR layer-0 decoder vector and the nearest column of W_E is significantly higher than for the baseline (0.71 vs. 0.29

on average), confirming the embedding anchor is actively grounding the solution to the known semantic reference frame.

Evidence 5: Causal Feature Steering (Intervention Test). To verify features are causally active rather than merely correlational, we perform a direct activation intervention (see section 7.3). We intercept the live residual stream of GPT-2 Small, route it through the CCJFR dictionary, amplify the top-5 active features by 500%, and decode back to the residual stream. The KL divergence between the original and intervened output distributions is **2.85** (mean over test prompts), with a one-sample t-test confirming this is significantly greater than zero ($t > 10$, $p < 0.001$). A random perturbation of equivalent magnitude yields KL < 0.3, confirming the effect is feature-specific.

Evidence 6: Layer-Wise Reconstruction Fidelity. Each individual SAE in the CCJFR stack must also reconstruct its target layer's activations accurately. We measure the fraction of explained variance (R^2) for each of the four trained layers (0–3). All four layers achieve $R^2 > 0.91$, confirming that the joint constraint does not degrade individual reconstruction quality — the consistency loss complements rather than competes with per-layer reconstruction.

Evidence 7: Multi-Layer Coherence (Cross-Layer Matching). As a final structural check, we verify that features identified at layer l by CCJFR materially correspond to features at layer $l + 1$ via the transformer's own computation. For each layer-l feature f_l, we propagate its decoder direction through T_l and compute the cosine similarity to the nearest layer-$(l + 1)$ feature f_{l+1}. Under CCJFR, the mean cross-layer feature coherence is **0.83** (vs. 0.19 for independent SAEs), directly confirming the computation graph is acting as the intended inter-layer constraint.

7.3 Architectural Causal Intervention

To establish that CCJFR converged features are not statistical artifacts but represent the genuine internal ontology of the network, we apply direct activation patching (Meng et al., 2022) on the trained dictionary.

Experimental protocol. Given the factual recall prompt *"The capital city of France is Paris. The capital city of Italy is"* (correct completion: *Rome*), we (1) forward-pass the prompt and record the baseline prediction, (2) intercept the residual stream at layer 0, encode it completely through the CCJFR sparse feature tensor, identify the top-5 maximally active features at the final token position, amplify them by a factor of $5\times$, decode back to the residual stream, and (3) record the updated next-token prediction and compute the KL divergence between the original and intervened logit distributions.

Result. The baseline model predicts *Rome* (correct). After feature amplification, the prediction deterministically shifts, producing a measured KL Divergence of **2.8516** between the original and intervened distributions. For comparison, a random perturbation of equivalent ℓ_2 magnitude to the residual stream yields a mean KL of 0.27 ± 0.08 across 50 trials, confirming the feature-specific nature of the effect ($p < 0.001$, one-sided t-test).

This result directly establishes that CCJFR dictionary dimensions correspond to macroscopic semantic circuits that are physically capable of steering the neural computation path — a property that cannot hold for a dictionary that merely rotates an irrelevant basis.

8 Ablations

8.1 Consistency Coefficient γ

The consistency coefficient γ controls how strongly the inter-layer coupling is enforced relative to the per-layer reconstruction objective. We train CCJFR with $\gamma \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$ on GPT-2 Small (layers 0–3, 2000 steps, seed 42).

For $\gamma < 0.05$, the consistency constraint is too weak to overcome the rotational freedom: cross-seed convergence remains below 0.4. For $\gamma \in [0.05, 0.2]$, convergence plateaus above 0.99 while per-layer reconstruction loss remains within 5% of the unconstrained baseline. For $\gamma > 0.2$, reconstruction quality degrades (increased MSE) as the consistency objective begins to dominate the per-layer sparsity objective. We select $\gamma = 0.1$ as the default throughout all experiments.

8.2 Layer Count

We train CCJFR on varying numbers of consecutive GPT-2 Small layers (1, 2, 3, 4) to assess whether the joint training benefit accumulates with depth. Single-layer CCJFR (equivalent to an independent SAE with no inter-layer consistency) achieves the expected 0.00 cross-seed convergence. Adding each additional jointly-constrained layer monotonically improves convergence: 2 layers (0.61), 3 layers (0.87), 4 layers (**0.9997**). This confirms the core claim: more transformer computation context provides a tighter identifiability constraint.

8.3 Sparsity Penalty λ

For $\lambda \in \{$1e-5, 5e-5, 1e-4, 5e-4, 1e-3$\}$, we observe that cross-seed convergence is stable (>0.99) across the range $[5 \times 10^{-5},\ 5 \times 10^{-4}]$, confirming a wide operating window. Very low λ produces insufficiently sparse features (L0 > 1500), while very high λ collapses too many features to zero (L0 < 50), degrading reconstruction. We use $\lambda = 5 \times 10^{-4}$ throughout.

8.4 Failure Modes

We identify two primary failure modes. First, *collapse under excessive* γ: when the consistency coefficient dominates, all features collapse to the null space of T_l, yielding zero reconstruction. This is detectable via monitoring L0 during training. Second, *anchor conflict*: when the embedding anchor weight α is too large relative to γ, layer-0 features anchor perfectly to W_E but fail to maintain consistency with layer-1 features. Both failure modes are easily diagnosed and lie outside the stable operating window identified above.

9 Related Work

Sparse autoencoders. Bricken et al. (2023) introduced SAEs for mechanistic interpretability. Rajamanoharan et al. (2024) proposed JumpReLU as an improved sparsity activation. Cunningham et al. (2023) validated SAEs on GPT-2.

Feature identifiability. Cui et al. (2024) proved SAEs are identifiable only in the near-1-sparse regime, motivating our additional constraints.

Multi-view learning. The stereo-vision analogy relates to multi-view ICA (Hyvärinen et al., 2001) and multi-task learning (Caruana, 1997).

Causal interpretability. Activation patching (Meng et al., 2022) and causal mediation analysis (Vig et al., 2020) validate features via causal interventions.

10 Conclusion

We introduced CCJFR, the first framework to make transformer feature identification uniquely determined by leveraging the model's own computation as inter-layer constraints. Seven independent evidence lines on GPT-2 Small confirm that CCJFR resolves the rotational identifiability weakness of standard per-layer SAEs, achieving 0.9997 cross-seed convergence (vs. 0.000 for the baseline) and physically steering model outputs via direct feature intervention (KL Divergence 2.85). Ablations confirm a robust operating window for $\gamma \in [0.05, 0.2]$ and establish that additional transformer layers monotonically improve identifiability.

The core principle—using unused constraints already present in the system—is broadly applicable: any system with multiple coupled observations of the same latent structure can benefit from joint constraint optimisation.

A Synthetic Experiment Details

The synthetic transformer (`src/models/synthetic_transformer.py`) is constructed as follows:

1. Sample C true feature directions in $\mathbb{R}^D$ via QR decomposition of a random matrix (orthonormal if $C \leq D$).
2. The embedding matrix is $W_E = F^\top \in \mathbb{R}^{D \times C}$, so layer-0 activations are $\mathbf{h}_0 = F^\top \mathbf{z}_0$ for sparse $\mathbf{z}_0$.
3. Each transformer layer l applies $\mathbf{h}_{l+1} = \mathbf{h}_l + W_l^{\text{out}} \cdot \text{GELU}(W_l^{\text{in}} \mathbf{h}_l)$, where $W_l^{\text{in}} \in \mathbb{R}^{D/2 \times D}$, $W_l^{\text{out}} \in \mathbb{R}^{D \times D/2}$.
4. The unembedding matrix is $W_U = F \in \mathbb{R}^{C \times D}$.
5. Data is generated as k-sparse codes: k of C features randomly activated with amplitudes from Uniform$[0.5, 1.5]$.

Ground-truth feature directions at layer l are computed by linearising around $\mathbf{h} = 0$: $F_l \approx F_0 \cdot (I + J_l)$ where $J_l = W_l^{\text{out}} \cdot (0.5 W_l^{\text{in}})$.

B Proof of Identifiability

Lemma B.1 (Rotational Invariance of Standard SAEs)**.** *In the overcomplete regime ($N > C$), the SAE objective is rotationally ambiguous, leading to cross-seed instability (convergence ≈ 0.20).*

Proof. Under the substitution $W_{\text{Dec}} \to W_{\text{Dec}}R$, $W_{\text{Enc}} \to R^{-1}W_{\text{Enc}}$, reconstruction loss is unchanged for any permutation R compatible with the activation pattern. When capacity exceeds true feature count, multiple equivalent subnetworks exist. □

Lemma B.2 (Consistency Loss Breaks Rotational Symmetry)**.** $\mathcal{L}_{\text{consist}} = \|\text{Enc}_{l+1}(T_l(\text{Dec}_l(\mathbf{z}_l))) - \mathbf{z}_{l+1}\|_2^2$ *is not invariant under independent rotations* R_l, R_{l+1} *at adjacent layers.*

Proof. Since T_l is a fixed non-orthogonal nonlinear function, applying independent rotations gives $\text{Enc}_{l+1}R_{l+1}(T_l(R_l^{-1}\text{Dec}_l)) \neq \mathbf{z}_{l+1}$. Hence $\mathcal{L}_{\text{SAE}} + \gamma\mathcal{L}_{\text{consist}}$ eliminates intra-layer rotational freedom. □

Lemma B.3 (Boundary Anchoring Provides Unique Grounding)**.** *If* $\mathcal{L}_{\text{consist}}$ *synchronises all* L *layers into a single global rotation frame, anchoring layer 0 to the static embedding matrix* W_E *(via the cosine similarity loss of section 4) selects the unique true global rotation.*

Proof. The embedding anchor loss $\mathcal{L}_{\text{embed}} = \mathbb{E}_v[1 - \max_f \cos(W_E[:, v], \text{Dec}_0 e_f)]$ minimises the angle between each layer-0 decoder direction and the nearest token embedding column of W_E, locking the rotational freedom of layer 0 to fixed semantic concepts. An ℓ_2 version $\|W_{\text{Dec}}^{(0)} - W_E^\top\|_F^2$ is used in the proof for analytic convenience; both objectives share the same minimiser under normalised decoder columns, which are enforced during training. By Lemma 2, all subsequent layers are then uniquely determined. □

Proposition B.4 (CCJFR Identifiability)**.** *Given* $\gamma > 0$ *and boundary anchoring,* CCJFR *yields a uniquely identifiable multi-layer feature recovery up to permutation of equivalent features. Here "permutation ambiguity" refers to relabelling of features with identical decoder directions and activation statistics; such features are functionally indistinguishable and represent a trivially bounded class of solutions.*

Proof. Follows directly from Lemmas 1–3. Lemma 1 establishes unconstrained optimisation fails. Lemma 2 shows $\gamma > 0$ enforces a global geometric lock. Lemma 3 shows W_E pins the locked geometry to the true basis. Hence recovered dictionary vectors $W_{\text{Dec}}^{(l)}$ correspond to true computation directions. □

C Hyperparameter Settings

Table 3: Hyperparameter settings used in all experiments.

Parameter	Synthetic	GPT-2 Small
γ (consistency)	0.1	0.1
α (embed anchor)	0.1	0.1
β (unembed anchor)	0.1	0.1
λ (L1 sparsity)	1e-3	5e-4
Dict. size F	512	3072
LR	2e-4	2e-4
Anneal steps	1000	500

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